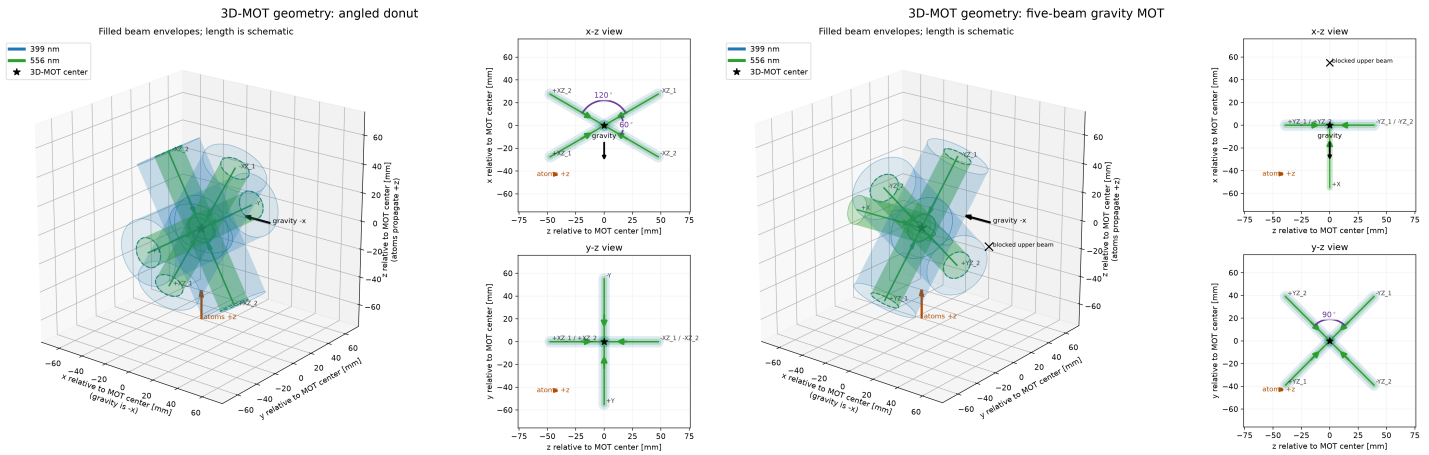


# Selecting the two 3D-MOT candidates

Simulation evidence before the full optimization campaign

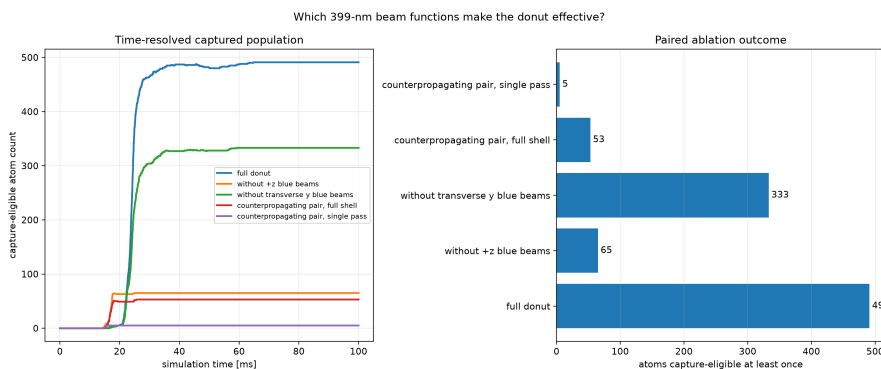
**Decision:** fully optimize the angled donut as the performance reference and the gravity-assisted five-beam MOT as the experimentally simpler alternative.

## 1. The candidates



## 2. Why reduced-blue six-beam variants were not selected

**Motivation.** Inspired by Plotkin's paper, we asked whether the six-blue-beam shell could be simplified to only two or four 399-nm beams. We therefore removed, retained, and repositioned blue-beam groups to measure the effect on capture.



### Paired 600-atom evidence

- **Full donut:** 491/600 usable (81.8%).
- **Remove +z blue pair:** 65/600 (10.8%).
- **Remove transverse y pair:** 333/600 (55.5%).
- **One blue pair, continuous shell:** 53/600 (8.8%).
- **One blue pair, single pass:** 5/600 (0.8%).

**Focused tuning helped, but did not close the gap:** 52/600 after intensity/detuning tuning, 115/600 for a refined entrance gate, and 188/600 (31.3%) for the best tested four-blue finite-gate endpoint.

**Interpretation:** transverse cooling, opposite longitudinal directions, and repeated blue-light access all matter. The evidence supports retaining the complete donut rather than spending the main search budget on reduced-blue variants.

# What the full optimization will deliver

High-level plan, reporting metric, and uncertainty statement

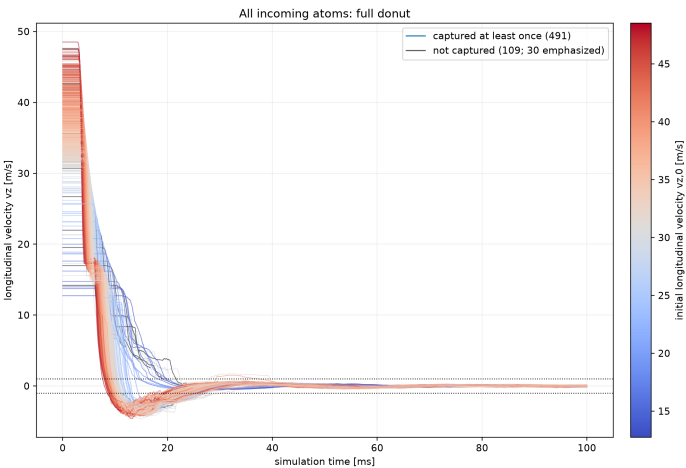
## Primary metric at 100 ms

An atom is usable when its distance from the MOT center is  $\leq 5$  mm and its total speed is  $\leq 1$  m/s. Earlier excursions do not permanently disqualify an atom.

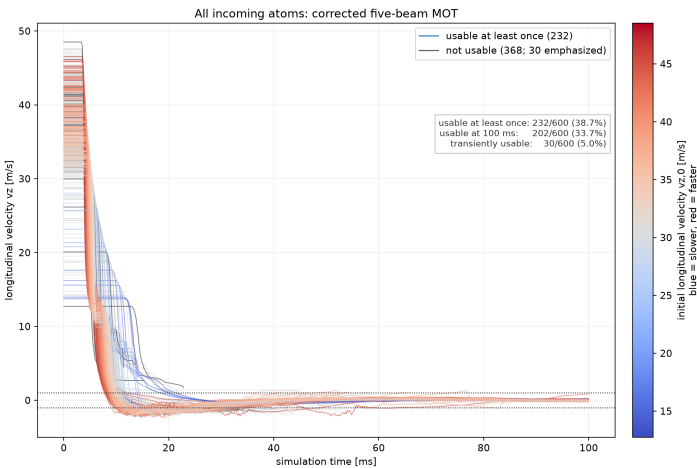
## Final result for each geometry

Recommended settings; conditional capture X +/- Y among 2D-MOT survivors; the corresponding usable atom number; and a near-optimal 7D or 10D parameter box whose performance loss is experimentally negligible.

## Longitudinal-velocity trajectories on the shared 600-atom input



**Donut:** broad entrance-velocity acceptance and stable slowing into the usable band.



**Five-beam:** the entrance acceptance is narrower, but the atoms usable at the end remain stable; the lower population is primarily a capture limitation, not post-capture loss.

## Optimized parameters and workflow

### 1 Donut (7)

Green beams:  $s_0$ , detuning, waist. Blue beams:  $s_0$ , detuning, waist. Plus magnetic-field gradient.

### 2 Five-beam (10)

Lower green:  $s_0$ , detuning, waist. Other four green: the same three. Four blue: the same three. Plus gradient.

### 3 Robust region

A local scan constructs the near-optimal hyperrectangle and checks its corners and interior.

### 4 Held-out prediction

New independent 2D-MOT survivor ensembles provide the final capture and flux estimate.

## How uncertainty is reported

X is the mean conditional capture fraction among 2D-MOT survivors, not among oven atoms. The expected 2D-MOT output is  $6.63701\text{e}9\text{--}6.92764\text{e}9$  atoms (order  $1\text{e}9$ ), so  $N3D = (X/100) \times N2D$ . The final Y is a 95% prediction half-width, using the larger of pooled counting uncertainty and ensemble/seed variation. Target:  $Y \leq 0.75$  percentage points.

## Laboratory decision after optimization

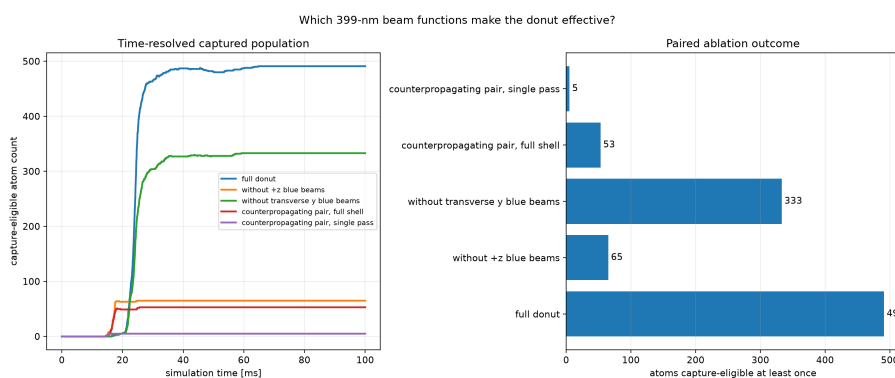
The donut is currently the clear atom-number reference. The five-beam MOT remains a candidate because fewer beams and easier alignment may justify a lower capture fraction. The final recommendation will present performance, robustness, power requirements, and implementation complexity side by side.

# Appendix A | Why the full donut is retained

Detailed evidence behind candidate selection

**Common comparison:** the same 600 validated 2D-MOT survivors, initial states, numerical settings, and recoil streams. Primary endpoint: usable at 100 ms (radius  $\leq 5$  mm and total speed  $\leq 1$  m/s). Earlier excursions do not disqualify atoms that return.

## Ablation results isolate the function of each blue-beam group

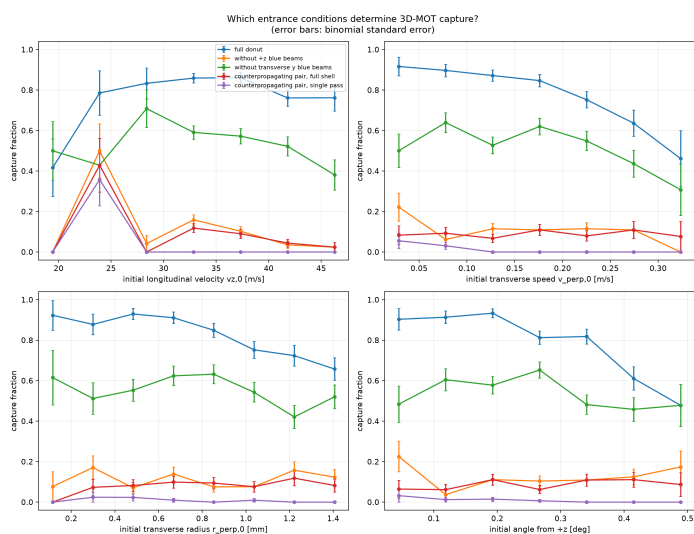
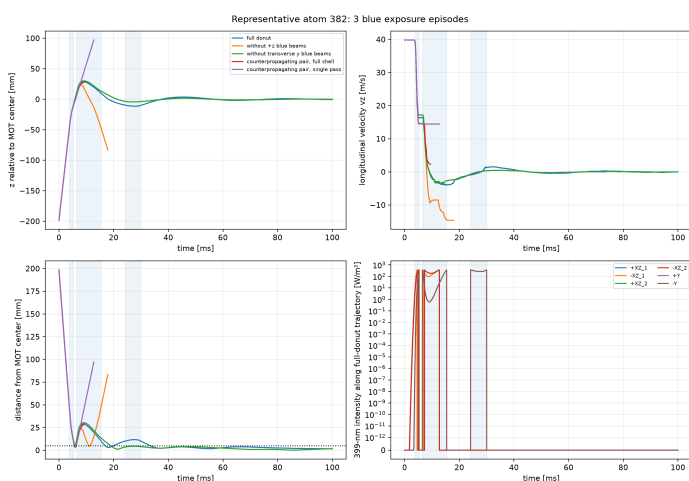


### Paired result at 100 ms

**Full donut:** 491/600 (81.8%)  
**No +z-component pair:** 65/600 (10.8%)  
**No transverse-y pair:** 333/600 (55.5%)  
**One pair, shell:** 53/600 (8.8%)  
**One pair, single pass:** 5/600 (0.8%)

**Physical reading.** The incoming cloud has transverse spread; atoms acquire 3D and oscillatory motion; some reverse their longitudinal direction; and successful atoms can encounter blue light more than once. Therefore beams that look redundant from the initial +z motion become cooling or restoring beams later.

## Trajectory and entrance-phase-space evidence



**Mechanism:** the representative atom has separated blue-exposure episodes; the population-level ablation is the stronger causal evidence.

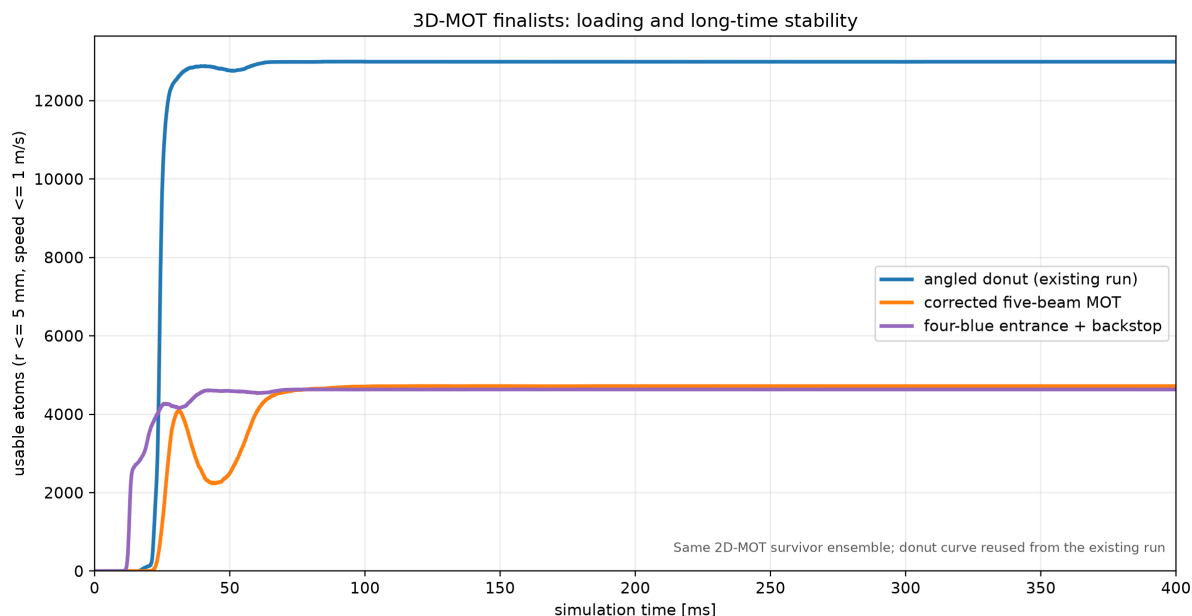
**Acceptance:** reduced-blue variants narrow the accepted initial phase space rather than merely reducing total optical power.

# Appendix B | Why the five-beam MOT remains

Corrected geometry, pre-optimization evidence, and practical tradeoff

**Architecture:** one green counterbeam on the gravity axis is removed and gravity assists the remaining unpaired green beam. The other four green beams form two counterpropagating pairs with blue shells. Force-sign tests include gravity and verify restoring behavior on both sides of  $x$ ,  $y$ , and  $z$ .

## Direct full-ensemble loading and stability through 400 ms



**15,840 shared survivors:** donut 12,992 at 100 ms and 12,989 at 400 ms; corrected five-beam 4,706 and 4,718; four-blue entrance + backstop 4,629 and 4,631. All three are effectively flat after loading, so no exponential lifetime is justified.

## Decision tradeoff

### Full angled donut

#### Advantages

- Highest capture by a large margin.
- Broad entrance-phase-space acceptance.
- Repeated cooling after reversals.
- Only 10 losses among 12,989 survivors from 100 to 400 ms in the tested hold.

#### Costs

- Six green and six blue beams.
- More optics, alignment, and access constraints.

### Gravity-assisted five-beam MOT

#### Advantages

- One fewer green MOT beam.
- Simpler laboratory construction and alignment.
- Repeatable nonzero capture in the corrected geometry.
- No measurable population decay through 400 ms at the tested point.

#### Risks

- Present capture is much lower.
- Gravity is fixed, not a velocity-selective optical counterbeam.
- Greater sensitivity to force balance, gradient, and the unpaired beam is plausible.

**Conclusion:** optimize both fairly. The donut is the performance reference; the five-beam remains because experimental simplicity may justify a lower atom number. Reduced-blue six-beam variants do not merit the main optimization budget.

# Appendix C | What is optimized

Variables, provisional domains, and laboratory constraints

## Seven-dimensional angled-donut search

### Six green core beams

1. s0
2. detuning [Gamma]
3. waist [m]

### Six blue shell beams

4. s0
5. detuning [Gamma]
6. waist [m]

### Magnetic field

7. gradient [G/cm]

## Ten-dimensional five-beam search

**Unpaired lower green:** 1-3. s0, detuning, waist

**Four paired green:** 4-6. s0, detuning, waist

**Four paired blue:** 7-9. s0, detuning, waist

**Magnetic field:** 10. gradient [G/cm]

Production code must support independent lower-green detuning and waist before this search starts.

## Planning domains - placeholders until laboratory approval

### Laser groups

Green s0: 0.5-40  
Green detuning: -35 to -5 Gamma  
Green waist: 5-20 mm  
Blue s0: 0.2-4  
Blue detuning: -6 to -0.5 Gamma  
Blue waist: 10-25 mm

### Gradients

Five-beam: 0.75-3 G/cm  
Donut: 0.5-6 G/cm

### Fixed geometry choices

Beam angles, polarizations, strong magnetic axis, cutoff geometry, MOT center, and gravity direction are not re-optimized.

### Why domains remain provisional

The five-beam winner reached the low-gradient boundary, so the final range must extend below 1.4 G/cm. Earlier donut settings also came from boundary studies; neither search should be centered too narrowly.

## Laboratory inputs required before production

1. Feasible minimum, maximum, adjustment resolution, calibration uncertainty, and drift for every s0, detuning, waist, and gradient.
2. Available optical power for each beam group. Since Gaussian-beam power scales approximately as  $s0 \times \text{waist}^2$ , s0 and waist cannot be optimized as an unconstrained rectangle.
3. Whether the unpaired green beam has independently controlled detuning and waist.
4. Whether beams within each group share common power, frequency, and waist controls.
5. The largest conditional-capture loss considered experimentally negligible for the robust parameter box.

**Implementation gate:** add per-group controls; verify force signs and exactly zero blue intensity in protected cores; reject infeasible power combinations; run 2-atom and 12-atom smoke tests; verify atomic checkpointing, restart, and live progress before allocating 24-hour jobs.

# Appendix D | Search execution

Multi-fidelity Optuna, independent validation, and Zeus execution

## Independent particle pools prevent optimistic reporting

**Existing production sample:** 20 independent 2D-MOT ensembles, 15,840 survivors total (756-840 per ensemble). Allocation: 12 ensembles for discovery, 4 for refinement and robust-box construction, and 4 for a preliminary held-out check. After all settings are locked, generate about 16,000 new survivors for final reporting and add ensembles adaptively if precision is insufficient.

## Search stages

### 1. Broad discovery

Common stratified 600-particle set per trial. Multivariate TPE. Minimum 350 valid donut trials and 600 valid five-beam trials. Continue if late improvement exceeds 1 point, optima sit on boundaries, or sampler seeds disagree.

### 2. Promotion

Promote strong or unresolved trials to a nested 1,800-particle set. The original 600 remain included, so comparison populations do not change.

### 3. Refinement

About 20 representatives per family: 3,000 particles, at least 3 recoil seeds, common particles and matched seeds. Rank conservatively; resolve paired differences to a 95% half-width  $\leq 0.4$  points or declare a tie.

### 4. Finalists

About five finalists per family on all 15,840 existing survivors with five recoil seeds. Favor robust, feasible, lower-power interior points over statistically indistinguishable numerical maxima.

## Zeus execution and restart safety

**Three long-lived workers:** 3 nodes x 200 cores, 64 GB per node, 24-hour walltime. Each node evaluates one complete 600-atom candidate at a time; three candidates run concurrently. Six hundred atoms still occupy all 200 workers in roughly three waves. Every trial is saved immediately with parameters, seeds, particle indices, counts, diagnostics, runtime, failure reason, and software revision, so the next 24-hour allocation resumes without losing completed trials.

# Appendix E | What the final claim means

Robust parameter box, uncertainty, flux propagation, and deliverables

## Near-optimal 7D or 10D hyperrectangle

**Goal:** a joint axis-aligned region, not independent one-dimensional error bars. Parameter interactions are retained. Proposed negligible-loss tolerance:  $\delta = 1.0$  percentage point of conditional capture, subject to laboratory approval.

**Validation design:** center; every one-parameter face midpoint; all cheap corners (128 for 7D, 1,024 for 10D); space-filling interior and face points; adversarial surrogate search for the worst predicted point; then high-statistics reevaluation of the worst corners and interior points. Use simultaneous, multiplicity-adjusted confidence bounds.

**Permitted wording:** within the declared domain, control resolution, tested corners, and adversarial validation design, the data provide 95% simultaneous confidence that no validated point loses more than  $\delta$ . This is a finite simulation guarantee, not a mathematical or experimental proof.

## Final prediction and uncertainty

### Conditional efficiency

X = mean usable 3D-MOT fraction among new, validated 2D-MOT survivors.

Y = 95% prediction half-width for a new equivalent ensemble, using the larger of pooled counting uncertainty and observed between-ensemble/recoil-seed variation.

Stopping target:  $Y \leq 0.75$  percentage points for each absolute efficiency.

### End-to-end atom flux

Modeled Yb-171 oven flux:  $7.38634 \times 10^{13}$  atoms/s.

Expected 2D-MOT flux:  $6.78232 \times 10^9$  atoms/s.

95% range:  $6.63701 \times 10^9$ – $6.92764 \times 10^9$  atoms/s.

For each family:  $N_{3D} = X \times N_{2D}$ , propagating uncertainty from both stages. Conditional 3D efficiency and total oven-to-3D efficiency are reported separately because their denominators differ.

## Final package delivered for each geometry

### Operating recommendation

- Nominal parameter values and required optical power.
- Conditional capture  $X \pm Y$  and predicted usable atom flux.
- Validated near-optimal hyperrectangle and declared  $\delta$ .

### Physics diagnostics

- Population versus time through 100 ms.
- Longer retention and  $\tau$  only when a genuine exponential tail passes loss and fit-quality checks.
- Final survivor checkpoints, common-scale phase-space acceptance, and representative captured and failed trajectories.

### Decision support

- Capture, robustness, power, beam count, alignment complexity, optical access, and implementation risk shown side by side. The simulation quantifies the tradeoff; the laboratory team decides how much atom number is worth the simpler geometry.